KDF1-Related Ectodermal Dysplasia (Ectodermal Dysplasia 12, ECTD12)

A Comprehensive Disease Characterization Report

Causal gene: *KDF1* (Keratinocyte Differentiation Factor 1). This report concerns the Mendelian ectodermal dysplasia caused by variants in ***KDF1*** (MONDO:0015024; OMIM phenotype #617337, "Ectodermal dysplasia 12, hypohidrotic/hair/tooth/nail type"). *KDF1* encodes keratinocyte differentiation factor 1, a cytoplasmic adaptor protein that governs the proliferation-versus-differentiation decision in epidermal/ectodermal progenitor cells. Because many ectodermal dysplasias are numbered serially and caused by different genes (e.g., ECTD1/EDA, ECTD10A/B/EDAR, ECTD11/EDARADD, ECTD2 & many others), this entry follows the identifiers above and specifically addresses the ***KDF1*** disorder, not the more common EDA/EDAR hypohidrotic ectodermal dysplasias.

> **Evidence base note.** This is an ultra-rare disorder described in a handful of families worldwide (first reported 2017). The report is compiled from primary human genetics case reports and from mouse/cell-model mechanistic studies. Evidence source types are flagged throughout as *[human clinical]*, *[mouse]*, *[in vitro/iPSC]*, or *[computational]*. Where disease-level data (epidemiology, survival, formal diagnostic criteria, trials) do not exist, this is stated explicitly.

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1. Disease Information

Overview. KDF1-related ectodermal dysplasia is a rare, predominantly **autosomal dominant** form of ectodermal dysplasia affecting derivatives of the embryonic ectoderm — **teeth, hair, nails, sweat glands, and skin**. The reported phenotype is a **hypohidrotic (hypohidrotic/hair/tooth/nail-type) ectodermal dysplasia** with tooth agenesis (hypodontia to anodontia), occasionally natal teeth, sparse hair, nail dystrophy, reduced sweating, and variable skin

involvement (palmoplantar keratoderma, atopic dermatitis). An allelic, mechanistically distinct **gain-of-function** phenotype combines ectodermal dysplasia with **hidradenitis suppurativa (HS)**.

Key identifiers. - **Gene:** KDF1 — HGNC:26624; **NCBI Gene ID 126695**; Ensembl **ENSG00000175707**; chromosome **1p36.11**; RefSeq transcript **NM_152365.3**; UniProt **Q8NAX2** (KDF1_HUMAN, 398 aa). Gene OMIM ***616758** (database-verified via mygene/UniProt/ ClinVar, Iteration 2).** - Disease OMIM: **#617337 — *Ectodermal dysplasia 12, hypohidrotic/ hair/tooth/nail type (ECTD12)*.** - MONDO: **MONDO:0015024.** - Orphanet: ORPHA:1810 **("Autosomal dominant hypohidrotic ectodermal dysplasia", MONDO:0015884) is the Orphanet grouping to which KDF1/ECTD12 is annotated (HPO/Monarch, verified Iteration 3); there is no separate ORPHA code specific to the KDF1 subtype.** - ICD-11: **EC20.0 / LD27.0 range (ectodermal dysplasia, anhidrotic/hypohidrotic);** ICD-10: **Q82.4 (Ectodermal dysplasia, anhidrotic). MeSH: "Ectodermal Dysplasia" (D004476); "Ectodermal Dysplasia 1, Anhidrotic" is a separate descriptor.** - MeSH gene concept:** KDF1.

Synonyms: ECTD12; ectodermal dysplasia 12, hypohidrotic/hair/tooth/nail type; ED-12; KDF1 ectodermal dysplasia syndrome; ectodermal dysplasia syndrome caused by mutation in KDF1.

Data source type: Disease-level knowledge is derived from **individual-patient case reports / family pedigrees** (not EHR or aggregated registries). No population-level or EHR-derived datasets exist for this disorder.

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2. Etiology

Disease causal factor — genetic (monogenic, Mendelian). The disease is caused by **heterozygous pathogenic variants in *KDF1***. The founding study mapped and exome-sequenced a multigenerational Saudi family and found the KDF1 variant **c.753C>G (p.Phe251Leu)** that fully co-segregated with autosomal dominant hypohidrotic ectodermal dysplasia *[human clinical]* (Shamseldin et al., 2017, P 27838789; variant–paper link confirmed via ClinVar VCV003655049):

> "We describe a multigenerational Saudi family with an autosomal dominant form of hypohidrotic ectodermal dysplasia in which positional mapping and exome sequencing identified a novel variant in KDF1 that fully segregates with the phenotype."

Genetic risk factors. - **Causal variants:** heterozygous missense variants in KDF1 (see §4). Both inherited (autosomal dominant) and **de novo** variants are reported (Manaspon et al., 2019, ; Pan et al., 2022, — de novo p.I304N). - **Modifier genes/susceptibility loci:** KDF1 sits in a TP63-centered regulatory network with IRF6, IKKα (CHUK), RIPK4, and stratifin (SFN); these are candidate genetic modifiers because they share the ectodermal knockout phenotype (Shamseldin 2017,). No formal GWAS/modifier study exists (ultra-rare disease).

P 30977908

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Environmental risk factors. None established. As a congenital Mendelian disorder, disease occurrence is determined by the germline variant, not by toxins, lifestyle, or exposures. Sex is not a determinant (autosomal, not X-linked, unlike the common EDA form). Family history is the principal non-molecular risk indicator.

Protective factors. No genetic or environmental protective factors are described. (By mechanism, a wild-type/corrected allele is "protective": CRISPR correction of the KDF1 mutation partially rescued cellular phenotypes in iPSC models — Peng 2026, *[in vitro]*.)

P 42267391

Gene–environment interactions. None documented. The disorder is essentially fully genetically determined; environmental contributions are limited to secondary care issues (e.g., hyperthermia risk from reduced sweating in hot environments).

3. Phenotypes

All phenotypes are **congenital / developmental** in origin, derived from ectodermal structures. Frequencies below are qualitative given the very small number of reported patients.

Phenotype	Type	Onset / course	Frequency (reported cohorts)	Suggested HPO term
Tooth agenesis — hypodontia, oligodontia, up to **anodontia**	Clinical sign / physical	Congenital (tooth development); apparent in childhood	Core, near-universal	HP:0000668 (Hypodontia); HP:0000677 (Oligodontia); HP:0009804 (Anodontia)
Natal / neonatal teeth	Clinical sign	Present at birth	Recurrent; Keramida 2025 is "third documented case of natal teeth"	HP:0000695 (Natal tooth)
Sparse hair / hypotrichosis	Physical manifestation	Congenital/ infancy	Common	HP:0001006 (Hypotrichosis); HP:0008070 (Sparse hair)
Hypohidrosis / reduced sweating	Clinical sign (lab-testable by sweat test)	Congenital; lifelong	Characteristic ("hypohidrotic" type)	HP:0000966 (Hypohidrosis)
Nail dystrophy (e.g., dystrophic toenails)	Physical manifestation	Childhood	Common	HP:0008404 (Nail dystrophy); HP:0008388 (abnormal toenail morphology)
Palmoplantar keratoderma / skin hyperkeratosis	Physical manifestation	Childhood	Reported	HP:0000982 (Palmoplantar keratoderma)
Atopic dermatitis / eczematous skin	Clinical sign	Childhood	Reported (Keramida 2025)	HP:0001047 (Atopic dermatitis)
Hidradenitis suppurativa /	Clinical sign			**HP:0040154 (Acne inversa)** /

Phenotype	Type	Onset / course	Frequency (reported cohorts)	Suggested HPO term
acne inversa (esp. gain-of-function allele)		Adolescence/ adulthood	Specific to GoF KDF1 variants	HP:0100680 (Hidradenitis suppurativa)

Authoritative HPO annotation set (HPO/Monarch for KDF1 / ECTD12, retrieved Iteration 3). Beyond the core features above, the curated HPO profile adds craniofacial, skin-appendage, and thermoregulatory terms. Full term list: - **Dentition:** Hypodontia **HP:0000668**; Natal tooth **HP:0000695**; Premature loss of primary teeth **HP:0006323**; Abnormal dental morphology **HP:0006482**; Abnormality of the dentition **HP:0000164**. - **Hair:** Sparse hair **HP:0008070**; Sparse lateral eyebrow **HP:0005338**; Sparse body hair **HP:0002231**; Scarring alopecia of scalp **HP:0004552**; Trichorrhexis nodosa **HP:0009886**. - **Sweat glands / thermoregulation:** Hypohidrosis **HP:0000966**; Hypoplastic sweat glands **HP:0007387**; **Malignant hyperthermia HP:0002047** (thermoregulatory vulnerability). - **Nails:** Nail dystrophy **HP:0008404**; Abnormal fingernail morphology **HP:0001231**; Subungual hyperkeratosis **HP:0008392**. - **Skin:** Eczematoid dermatitis **HP:0000964**; Keratosis pilaris **HP:0032152**; Orthokeratosis **HP:0040162**; Thin skin **HP:0000963**; Abnormal skin pigmentation **HP:0001000**; Palmar hyperlinearity **HP:0033252**; Knuckle pad **HP:0032541**. - **Craniofacial:** Prominent forehead **HP:0011220**; Depressed/Concave nasal ridge **HP:0000457 / HP:0011120**; Short philtrum **HP:0000322**; Thick vermilion border **HP:0012471**; **Cleft palate HP:0000175**. - **Inheritance:** Autosomal dominant inheritance **HP:0000006**.

Note: *Malignant hyperthermia* (HP:0002047) in the HPO profile likely reflects heat-intolerance/ hyperthermia risk secondary to defective sweating rather than classic anesthetic malignant hyperthermia; this distinction should be clarified clinically. *Cleft palate* and the craniofacial dysmorphism broaden the syndrome beyond the simple hair/tooth/nail/sweat tetrad.

Keramida et al. 2025 (P 40463401) summarize the carrier-family manifestations *[human clinical]*:

> "Clinical manifestations of the carrier family members included natal teeth at birth, hypodontia, atopic dermatitis and dystrophic toenails."

Severity & progression. Variable expressivity even within a family; the dental phenotype ranges from mild hypodontia to severe non-syndromic anodontia (Pan 2022, P 35641834).

Structural ectodermal defects are **congenital and stable** (non-progressive), though sequelae (dental, thermoregulatory) are lifelong. Hidradenitis suppurativa in GoF cases is a chronic, relapsing inflammatory course.

Quality-of-life impact. No formal EQ-5D/SF-36 data for this specific disorder. Expected impacts (by analogy to ectodermal dysplasias): impaired mastication/nutrition and speech and psychosocial burden from tooth agenesis; heat intolerance from hypohidrosis; cosmetic/psychosocial impact of sparse hair and nail dystrophy; substantial morbidity and pain where hidradenitis suppurativa is present.

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4. Genetic / Molecular Information

Causal gene. *KDF1* (Keratinocyte Differentiation Factor 1), 1p36.11; HGNC:26624; NCBI Gene ID **126695**; Ensembl ENSG00000175707; transcript NM_152365.3; protein **UniProt Q8NAX2** (398 aa, cytoplasmic/cell-junction). Mouse ortholog *Kdf1* (NCBI Gene **69073**; MGI:1916323; historically "1810019J16Rik"; Lee 2013, P 24075906).

Pathogenic variants (human). Reported variants are **heterozygous missense** substitutions that cluster in a **C-terminal functional hotspot (~residues 251–304)** of the 398-aa protein: - **c.753C>G, p.Phe251Leu** — ClinVar **Pathogenic**; this is the **original index Saudi autosomal-dominant HED family variant** (ClinVar VCV cites Shamseldin 2017, P 27838789). - **c.760C>T, p.His254Tyr** — ClinVar **Likely pathogenic**; ClinVar VCV cites the **gain-of-function hidradenitis-suppurativa report** (Ziegler 2023, P 37144643) — an IKKα-stabilizing GoF allele. - **c.795C>A, p.Asp265Glu** — ClinVar **Likely pathogenic**, Ectodermal dysplasia 12. - **c.812A>C, p.Lys271Thr (p.K271T)** — autosomal dominant ED-12, 3-generation Greek family (Keramida 2025,

). P 40463401

- **c.908G>C (reported c.G908C), p.Arg303Pro (p.R303P)** — co-segregating in non-syndromic tooth agenesis (Zeng 2019,). P 30384154

- **c.911T>A, p.Ile304Asn (p.I304N)** — **de novo**, severe non-syndromic anodontia (Pan 2022,). P 35641834

- Additional variants in the index AD family (Shamseldin 2017,) P 27838789 and a de novo variant (Manaspon 2019,) P 30977908 substantiating KDF1 causality. - **Gain-of-function variants** that stabilize IKKα cause HS + ectodermal dysplasia (Ziegler et al., 2023,). P 37144643

Variant classification (ACMG/AMP) — ClinVar landscape (verified Iteration 2, ~60 KDF1 records). The gene is **VUS-predominant**: ~47 Uncertain significance, 8 Likely benign, 2 Likely pathogenic, 1 Pathogenic. Confirmed P/LP variants are assigned to **"Ectodermal dysplasia 12, hypohidrotic/hair/tooth/nail type."** Classification relies heavily on co-segregation, de novo occurrence, and functional assays because population data are sparse. This VUS-heavy distribution reflects the gene's recent discovery (2017) and rarity.

Variant type/class. Predominantly **missense**. (No large structural/CNV mechanism is established for this disorder.)

Allele frequency. Causal variants are **absent or ultra-rare** in gnomAD/1000 Genomes/ExAC, consistent with pathogenicity; no common susceptibility allele is known.

Somatic vs germline. Germline (constitutional) — inherited or de novo. No somatic/cancer role established for this disease.

Functional consequences. Two mechanistically opposite classes converge on the same pathway: - **Loss-of-function-type / hypomorphic** missense variants impair KDF1's pro-differentiation function → ectodermal dysplasia and tooth agenesis. In vitro, mutant KDF1 shows altered mRNA/protein expression (Pan 2022,

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"In vitro functional studies showed altered mRNA and protein expression levels of the mutant KDF1." - **Gain-of-function** variants **stabilize IκB kinase α (IKKα)** causing HS + ED (Ziegler 2023,

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P 37144643

Population constraint & mechanism inference (gnomAD, verified Iteration 3). KDF1 is **moderately loss-of-function constrained**: pLI = 0.83, **LOEUF = 0.58**, observed/expected LoF = 0.35 (11 observed vs 31.6 expected; lof_z = 3.11). Missense constraint is only mild (oe_mis = 0.87, mis_z = 1.33). Because (i) heterozygous LoF alleles do occur in population controls, (ii) **all reported pathogenic alleles are missense** clustered at aa ~251–304, and (iii) a defined gain-of-function (IKKα-stabilizing) class exists, the dominant disease mechanism most likely reflects a **dominant-negative or gain-of-function missense effect rather than simple haploinsufficiency** *[computational/population-genetics inference]*. This is consistent with the recessive (homozygous-null) Kdf1 mouse, implying human heterozygous null may be largely subclinical.

Allele frequency. Pathogenic alleles are absent/ultra-rare in gnomAD; the gene tolerates some common missense variation (476 observed missense), so classification of novel missense variants requires segregation/functional support.

Modifier genes. Candidate modifiers are network partners TP63, CHUK/IKKα, IRF6, RIPK4, SFN (stratifin) (Shamseldin 2017). No quantitative modifier data.

Epigenetic information / chromosomal abnormalities. None described for this disorder; no methylation signature or cytogenetic abnormality is implicated. Disease is caused by point variants in a single gene.

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5. Environmental Information

- **Environmental factors / toxins / radiation / occupational exposures:** none implicated — this is a monogenic congenital disorder.
- **Lifestyle factors:** none causal. Relevant only to management (heat avoidance due to hypohidrosis; smoking/obesity are known aggravators of hidradenitis suppurativa in

general and would be reasonable to address in GoF-variant patients, though not studied specifically here).

- **Infectious agents:** none. Secondary skin infection can complicate hidradenitis suppurativa lesions but is not a cause of the disorder.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous germline *KDF1* missense variant** (inherited AD or de novo) alters KDF1 protein function in ectodermal progenitor/keratinocyte lineages. *[human clinical + in vitro]*
2. **Loss-of-function branch:** reduced/altered KDF1 activity **leads to** failure to restrain progenitor proliferation (loss of KDF1-mediated **inhibition of p63) and** failure to deliver the pro-differentiation signal. *[mouse]* (Lee 2013)
3. Mechanistically, impaired KDF1 **results in** failure to recruit USP7 and **deubiquitinate/stabilize IKKα**, so IKKα-dependent terminal differentiation signaling is not sustained. *[mouse + in vitro]* (Li 2020)
4. The disturbed **proliferation⇄differentiation balance** in epidermal/ectodermal appendage placodes **leads to** defective morphogenesis of teeth, hair follicles, nails, and eccrine sweat glands. *[mouse, inferred for human]*
5. This **results in** the clinical phenotype: tooth agenesis/natal teeth, hypotrichosis, nail dystrophy, hypohidrosis, and skin keratinization abnormalities. *[human clinical]*

Gain-of-function branch (parallel): IKKα-**stabilizing** *KDF1* variants **lead to** excess/dysregulated IKKα→NF-κB-pathway activity in follicular/apocrine-gland epithelium, **resulting in** hidradenitis suppurativa combined with ectodermal dysplasia. *[human clinical]* (Ziegler 2023)

1. A newly proposed upstream/parallel contributor (from iPSC modeling): the *KDF1* mutation **disrupts synaptic structure/signaling and raises excitatory neurotransmitter (NMDA-type) tone**, which may **impede ectodermal organogenesis**; NMDA-receptor antagonism (MK-801) and CRISPR correction partially rescue the defect. *[in vitro/iPSC — inferred, not yet demonstrated in vivo human]* (Peng 2026)

Supporting detail by category

- **Molecular pathways: p63 regulatory network** (KDF1 inhibits p63-driven proliferation); **IKKα / NF-κB (IκB-kinase) signaling** via KDF1–USP7-mediated deubiquitination; the TP63–IRF6–IKKα–RIPK4–stratifin ectodermal-differentiation module. Suggested pathway terms: NF-κB signaling (KEGG/Reactome), keratinization (Reactome R-HSA-6805567).
- **Cellular processes:** regulation of cell proliferation (GO:0042127), keratinocyte/epidermal cell differentiation (GO:0030216), protein deubiquitination (GO:0016579), ectoderm/epidermis development (GO:0007398, GO:0008544), epithelial placode morphogenesis.
- **Protein dysfunction:** KDF1 (398 aa) is a cytoplasmic/cell-junction adaptor-scaffold; pathogenic missense variants cluster in a **C-terminal hotspot (~aa 251–304)** near functional interaction surfaces, impairing or distorting partner binding (USP7, IKKα). Population constraint data (gnomAD: LoF-depleted but missense-tolerant; §4) plus exclusively-missense pathogenic alleles indicate a **dominant-negative** mechanism for the ED phenotype and a **gain-of-function** mechanism (abnormal IKKα stabilization) for the HS phenotype — i.e., altered protein function/interaction rather than mere loss of dosage. Suggested structural resources: UniProt Q8NAX2; AlphaFold model AF-Q8NAX2-F1. UniProt annotates KDF1 as cytoplasmic and cell-junction-localized, "Required both as an inhibitor of basal cell proliferation and a promoter of differentiation of basal progenitor cell progeny."
- **Immune-system involvement:** via **IKKα/NF-κB**, dysregulation produces sterile/chronic inflammatory skin disease (hidradenitis suppurativa) in GoF cases.
- **Tissue damage mechanism:** primarily **developmental (hypoplasia/agenesis)** rather than degenerative — appendages fail to form normally; in HS, chronic follicular occlusion/inflammation drives tissue damage.
- **Biochemical abnormality:** aberrant ubiquitin–proteasome handling of IKKα (loss of USP7-dependent stabilization, or pathological over-stabilization).
- **Molecular profiling:** single-cell RNA-seq of patient iPSC-derived embryoid bodies showed a **shortage of specific cell populations related to synapse structure/signaling** (Peng 2026, P 42267391) *[in vitro]*.
- **Functional genomics:** KDF1 was discovered by an **unbiased forward-genetic (ENU) screen** in mouse (Lee 2013); CRISPR correction rescues iPSC phenotypes (Peng 2026).

Upstream vs downstream: the KDF1 variant is upstream; p63 de-repression and IKKα destabilization are proximal downstream nodes; appendage dysmorphogenesis and the clinical phenotype are terminal. **Cell types:** keratinocytes (CL:0000312), epidermal basal/progenitor cells (CL:0002187). **GO biological processes:** GO:0030216, GO:0008544, GO:0042127, GO:0016579.

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7. Anatomical Structures Affected

- **Organ/system level: integumentary system** is primary — **skin/epidermis** (UBERON:0001003), **hair follicles** (UBERON:0002073), **nails** (UBERON:0001705), **eccrine sweat glands** (UBERON:0002031 / sweat gland UBERON:0001820); **teeth / dentition** (UBERON:0001091). Secondary: apocrine-gland–bearing skin (axillae, groin) in HS. Thermoregulation is impaired secondary to sweat-gland hypoplasia.
- **Tissue/cell level: stratified squamous epithelium (epidermis)** and ectodermal-appendage epithelia; affected cells are **keratinocytes** (CL:0000312) and **epidermal progenitor/basal cells** (CL:0002187). Dental epithelium (enamel organ) is affected during odontogenesis.
- **Subcellular level: cytoplasm** (KDF1 is cytoplasmic; GO:0005737), with functional impact on **ubiquitin–proteasome machinery** and signaling to the **nucleus** (NF-κB). No mitochondrial/lysosomal involvement implicated.
- **Localization / lateralization:** manifestations are **bilateral and generally symmetric** (congenital developmental distribution); HS affects intertriginous (apocrine) sites bilaterally.

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8. Temporal Development

- **Onset: congenital / developmental.** Dental and appendage defects originate during embryonic ectodermal development; natal teeth are present at birth; hypotrichosis, hypohidrosis, and nail changes manifest in infancy/early childhood. HS (GoF allele) typically emerges in adolescence/adulthood.
- **Onset pattern:** insidious/congenital (structural features), with HS following a chronic relapsing course.

- **Progression:** the **ectodermal structural defects are non-progressive/stable** (fixed developmental deficits), lifelong. No "staging" system applies. HS is chronic, relapsing–remitting, and can progress through Hurley stages where present.
- **Disease duration:** chronic, lifelong.
- **Remission:** structural features do not remit; they are managed symptomatically. **Critical window:** the key opportunity for intervention is developmental/early childhood for dental and functional management; the iPSC model raises a speculative prenatal/early neuro-developmental window (NMDA signaling) *[in vitro, unproven clinically]*.

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9. Inheritance and Population

- **Inheritance pattern: Autosomal dominant** (OMIM #617337), with documented **de novo** occurrence. Gene is autosomal (1p36.11) — contrasting with the common X-linked EDA hypohidrotic ED.
- **Penetrance / expressivity:** co-segregation in multigenerational families indicates high penetrance; **expressivity is variable** (ranging from isolated/non-syndromic tooth agenesis to full syndromic ED, even within the same gene/region).
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity:** not specifically established; the index family was consanguineous Saudi (Shamseldin 2017) but AD transmission is the rule. No founder variant reported.
- **Carrier frequency:** not applicable in the recessive sense (dominant disorder); causal alleles are ultra-rare in gnomAD.
- **Epidemiology: No prevalence/incidence estimates exist** — the disorder is known from only a few families (Saudi Arabia, China, Greece, and others). It is **ultra-rare**. Ectodermal dysplasias collectively affect ~1 in 10,000–100,000, but KDF1 accounts for a very small fraction.
- **Population demographics / geography:** reported across multiple ethnicities (Middle Eastern, East Asian, European); no demonstrated ethnic predilection or endemic region. **Sex ratio:** expected ~1:1 (autosomal). Age distribution: present from birth, recognized across all ages.

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10. Diagnostics

- **Clinical recognition:** diagnosis is suspected on the clinical triad/tetrad of **tooth agenesis (± natal teeth), sparse hair, nail dystrophy, and hypohidrosis**, with or without skin keratoderma; HS + ED should prompt KDF1 testing for the GoF allele.
- **Genetic testing (confirmatory, the diagnostic gold standard):**
- **Whole-exome sequencing (WES)** — the method by which KDF1 was discovered and by which most cases are diagnosed (Shamseldin 2017; Keramida 2025; Pan 2022).
- **Multigene ectodermal-dysplasia / tooth-agenesis panels** including KDF1.
- **Single-gene (KDF1) sequencing** and **Sanger confirmation/segregation** testing of relatives.
- **Whole-genome sequencing** useful when panels/WES are negative. CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not indicated** (point-variant, nuclear, autosomal disease).
- **Supporting tests:** dental **panoramic radiography (orthopantomogram)** to document agenesis; **sweat/pilocarpine or starch-iodine sweat test** for hypohidrosis; **skin/dental biopsy** can show the differentiation defect but is not routine. Immunohistochemistry of KDF1 in developing tooth germs supports biological plausibility (Zeng 2019).
- **Omics-based diagnostics:** research-level only (scRNA-seq, functional expression assays). Not used clinically.
- **Biomarkers:** no circulating biomarker; the molecular variant is the biomarker.
- **Clinical criteria:** no disorder-specific standardized diagnostic criteria; diagnosis = compatible phenotype + pathogenic KDF1 variant.
- **Differential diagnosis:** other hypohidrotic ectodermal dysplasias — **EDA (X-linked), EDAR, EDARADD** (XLHED/HED); **WNT10A**-related ED/tooth agenesis; **TP63**-related syndromes (EEC, AEC, Hay–Wells); **IRF6** (Van der Woude/popliteal pterygium); isolated non-syndromic **tooth agenesis** genes (PAX9, MSX1, AXIN2, LRP6). KDF1 is distinguished by autosomal dominant inheritance plus the specific KDF1 variant and the natal-teeth/HS associations.
- **Screening: cascade genetic testing** of at-risk relatives once a familial variant is identified; prenatal/preimplantation testing is technically possible for known familial variants.

11. Outcome / Prognosis

- **Survival/mortality: normal life expectancy**; this is **not a life-limiting disorder**. No disease-specific mortality. (Caveat: untreated severe hypohidrosis carries a theoretical risk of hyperthermia, well-managed by environmental measures.)
- **Morbidity/function:** morbidity is driven by **dental disability** (mastication, nutrition, speech, aesthetics), **heat intolerance**, **cosmetic/psychosocial** impact (hair, nails), and — in GoF cases — **significant pain, scarring, and disability from hidradenitis suppurativa**.
- **Disease course:** structural defects are stable/lifelong; HS is chronic-relapsing. **Recovery:** structural deficits do not spontaneously resolve but are highly amenable to prosthetic/surgical rehabilitation.
- **Quality-of-life measures:** no disorder-specific instruments reported; generic tools (EQ-5D, SF-36, PROMIS, Oral Health Impact Profile, and DLQI/HS-specific indices for HS) would be applicable.
- **Prognostic factors:** severity of tooth agenesis (hypodontia vs anodontia) and presence/absence of HS are the main determinants of functional burden; genotype (LoF-type vs IKKα-stabilizing GoF) predicts the clinical picture.

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12. Treatment

There is no disease-modifying or gene-specific therapy; management is supportive, symptomatic, and multidisciplinary. No approved drug or trial (ClinicalTrials.gov) is specific to KDF1-ED.

- **Dental / oral rehabilitation (central):** pediatric dentures, overdentures, **dental implants** and prosthodontics, orthodontics; management of natal teeth (monitoring vs extraction if aspiration/feeding risk). *NCIT suggestions:* Dental Prosthesis (C50111), Dental Implant (C62701), Orthodontic procedures.
- **Thermoregulation/hypohidrosis:** behavioral cooling, hydration, avoidance of heat/exertion; environmental control.
- **Skin/hair/nails:** emollients and keratolytics for keratoderma; topical care for atopic dermatitis; cosmetic measures/wigs for hypotrichosis; nail care.
- **Hidradenitis suppurativa (GoF cases):** standard HS management — topical/systemic antibiotics, anti-inflammatory therapy, **TNF-α inhibitors (e.g., adalimumab)** and other

biologics, and surgical excision of chronic lesions. *NCIT:* Adalimumab (C1876). Given the IKKα/NF-κB mechanism, NF-κB-pathway-directed anti-inflammatory strategies are mechanistically rational (research-level).

- **Pharmacogenomics:** none specific.
- **Advanced/experimental therapeutics:** no gene, cell, or RNA therapy in clinical use. **Proof-of-concept** rescue by **CRISPR correction of the KDF1 variant** and by **NMDA-receptor antagonist MK-801** in patient iPSC models (Peng 2026, P 42267391) *[in vitro only; not a clinical therapy]*.
- **Supportive/rehabilitative:** speech therapy, nutritional support, psychological support.
- **Treatment strategy:** individualized, lifelong multidisciplinary care (dentistry/prosthodontics, dermatology, genetics, and — for HS — dermatologic surgery).

13. Prevention

- **Primary prevention:** not possible for a germline Mendelian disorder; **genetic counseling** and **reproductive options** (prenatal diagnosis, preimplantation genetic testing for a known familial variant) are the principal preventive measures (autosomal dominant → 50% transmission risk from an affected parent).
- **Secondary prevention / early detection: cascade genetic testing** of relatives; early dental assessment and intervention; early recognition of natal teeth. Newborn screening does not include KDF1.
- **Tertiary prevention:** prevent complications — maintain hydration/cooling to avoid hyperthermia; early prosthodontic rehabilitation to preserve nutrition/oral function; early HS management to limit scarring/disability.
- **Counseling:** genetic counseling regarding autosomal dominant inheritance, variable expressivity, and de novo occurrence is central (per ACMG/NSGC practice).
- **Immunization / public-health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** mouse *Kdf1* (originally **1810019J16Rik**; NCBI Gene **69073**; MGI:1916323; chromosome 4), highly conserved with human KDF1 (ENSG00000175707). *[mouse]*
- **Natural disease in other species:** no naturally occurring companion-animal or wildlife KDF1 disease is catalogued (OMIA); the disorder is defined in humans and modeled experimentally in mice. Veterinary relevance is limited to experimental models.
- **Comparative biology:** the **Kdf1-/- mouse recapitulates the human ectodermal phenotype**, indicating strong evolutionary conservation of KDF1's role in the proliferation–differentiation balance (Shamseldin 2017; Lee 2013). Comparative pathology shows epidermal hyperproliferation with failed terminal differentiation in mouse mirroring human ectodermal appendage defects.
- **Transmission:** not applicable (non-communicable; no zoonotic potential).

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15. Model Organisms

- **Mouse (primary model): Kdf1 knockout / ENU-induced loss-of-function mutant** (Lee, Kong, Weatherbee 2013,

P 24075906

) — discovered KDF1 via forward genetics *[mouse]*:

> "we identified an essential regulator of the proliferation-differentiation balance in epidermal progenitor cells, the Keratinocyte differentiation factor 1 (Kdf1 …) gene … Kdf1 mutants display both uncontrolled cell proliferation in the epidermis and failure to develop terminal fates."

- **Model type:** mammalian germline KO/point mutant; conditional/knock-in models feasible.
- **Phenotype recapitulation: high** — the Kdf1-/- mouse reproduces the human ectodermal phenotype (basis for human causality inference; Shamseldin 2017). Also used to dissect the KDF1–USP7–IKKα mechanism (Li et al. 2020,

P 32239614

).

- **Limitations:** rodent skin/appendage biology differs from human dentition; dominant human missense (incl. GoF) effects are not fully captured by simple knockouts.
- **Human iPSC / organoid models (in vitro):** patient-derived iPSCs, embryoid bodies, epidermal-progenitor and neuronal differentiation; scRNA-seq; **CRISPR-corrected isogenic controls** (Peng et al. 2026,

  P 42267391

)
  [in vitro]. Applications: mechanism (synaptic/NMDA signaling in ectodermal organogenesis), drug testing (MK-801), and genotype–phenotype dissection. Limitation: in-vitro system; organogenesis only partially modeled.
- **In vitro cell assays:** HEK/keratinocyte overexpression assays comparing wild-type vs mutant KDF1 mRNA/protein levels and interactions (Pan 2022; Li 2020).
- **Resources:** MGI (mouse Kdf1), IMPC/KOMP (knockout resources), Cellosaurus/ATCC (iPSC lines in cited studies).

Evidence Summary (key PMIDs)

PMID	Year	Contribution	Evidence type
27838789	2017	KDF1 identified as cause of AD hypohidrotic ED (index Saudi family, variant c.753C>G/p.Phe251Leu); mouse recapitulation	Human + mouse
24075906	2013	Kdf1 forward-genetic discovery; proliferation–differentiation balance; p63 inhibition	Mouse
32239614	2020	KDF1–USP7 deubiquitinates/stabilizes IKKα to drive epidermal differentiation	Mouse + in vitro
30384154	2019	KDF1 c.G908C p.R303P in non-syndromic tooth agenesis (phenotypic heterogeneity)	Human
35641834	2022	De novo c.911T>A p.I304N causing non-syndromic anodontia; functional assays	Human + in vitro
30977908	2019	De novo KDF1 variant substantiating ED causality	Human
37144643	2023	Gain-of-function KDF1 variants → hidradenitis suppurativa + ED via IKKα stabilization	Human
40463401	2025	Novel c.812A>C p.K271T in 3-generation ED-12 family; natal teeth, hypodontia, dystrophic nails, atopic dermatitis	Human
42267391	2026	iPSC model: synaptic/NMDA dysregulation impairs ectodermal organogenesis; CRISPR/MK-801 rescue	In vitro/iPSC

Supported vs Refuted / Open Hypotheses

- **Supported:** (1) KDF1 heterozygous variants cause autosomal dominant ectodermal dysplasia (ECTD12). (2) KDF1 controls the epidermal proliferation–differentiation decision via p63 inhibition and USP7-mediated IKKα stabilization. (3) A gain-of-function, IKKα-stabilizing allele produces an HS + ED phenotype. (4) The phenotypic spectrum extends to non-syndromic tooth agenesis/anodontia (variable expressivity).

- **Open / not established:** precise penetrance estimates; whether the iPSC-derived "neuronal/NMDA" mechanism operates in vivo in humans; existence of modifier alleles; any natural-disease counterpart in other species; disorder-specific epidemiology, QoL, and prognostic datasets.

Limitations

Evidence rests on a small number of families and model systems; no population-level epidemiology, controlled trials, or formal natural-history studies exist. Core identifiers (Gene 126695, HGNC:26624, UniProt Q8NAX2, Ensembl ENSG00000175707, NM_152365.3, gene OMIM *616758) and ClinVar P/LP variants were database-verified (Iteration 2); the disease phenotype MIM (#617337) and the index Saudi-family variant (c.753C>G/p.Phe251Leu) were both resolved via HPO/Monarch and ClinVar–PMID cross-linking (Iterations 3–4). Frequencies remain qualitative owing to the very small patient numbers; per-phenotype penetrance percentages cannot be reliably derived from the few reported families.

Future directions

Expand genotype–phenotype correlation (LoF vs GoF); structural characterization of KDF1 (UniProt Q8NAX2 / AlphaFold AF-Q8NAX2-F1) to map variant effects on USP7/IKKα binding; registry building for epidemiology and natural history; test NF-κB/IKKα-directed therapy in GoF HS cases; validate the iPSC-derived synaptic-signaling mechanism in vivo.